MIL-C-81344(AS)
AMENDMENT-1
1 MAY 1968

MILITARY SPECIFICATION

CONVERTER, SIGNAL DATA CV-2461( )/A

This amendment forms a part of Military Specification MIL-S-81344(AS) dated 27 Dec 1965 and has been approved by the Naval Air Systems Command; Department of the Navy.

Page 1, Specification Title: Delete and substitute the title of this amendment.

Page 1, paragraph 1.1: Change "11 bit binary" to "12-14 bit binary".

Page 1, paragraph 3.2(1): Change "WR-101, part 2" to "WR-101, part II", and add: "except for linear microelectronic circuits".

Page 2, paragraph 3.2.1: Add: "except that microelectronic devices shall be in accordance with WR-101, part II".

Page 2, paragraph 3.2.2: Delete

Page 2, paragraph 3.2.3, line 2 after "Assemblies shall": Delete and substitute "meet the requirements of AR-5".

Page 2, Add:

"3.2.4 Maintenance Module - A maintenance module is defined as the least complex segment of the contract end article(s) which is capable of being removed and replaced at organizational level maintenance using tools, support equipment and procedures approved for the end article(s). Further, it is the segment to which the Contractor will provide fault detection and isolation techniques consistent with scheduled and unscheduled organizational level maintenance requirements.

3.2.5 Microelectronic Interchangeability - The interchangeability provisions of AR-5 shall govern microelectronic interchangeability".

Page 2, paragraph 3.3.1: Delete "50 pounds" and substitute 80.0 pounds".

Page 2, paragraph 3.3.2.3.1, line 1: Delete "specified or" and on line 2 Delete "specified".

Page 2, paragraph 3.3.2.3.2: Delete "MTBF To Be Demonstrated By Test" and substitute "Specified MTBF".

FSC 6125

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Page 2, Add:

"3.3.2.3.3 Reliability Program - The contractor shall establish and conduct a reliability assurance program in accordance with MIL-STD-785".

Page 3, paragraph 3.3.2.4: Delete "Class 1" and substitute "Class 1A".

Page 3, paragraph 3.3.3, line 11, after "cases" add "of failure".

Page 3, paragraphs 3.3.3.1, 3.3.3.1.2, 3.3.3.1.3, and 3.3.3.1.4: Delete and substitute:

"3.3.3.1 Maintainability Program - The contractor shall establish and conduct a maintainability program in accordance with MIL-STD-470 and MIL-STD-471. All maintenance actions shall be capable of being performed by a technician with a comprehensive ability level as follows:

(a) Civilian Education - High school graduate or equivalent.
(b) Technical Training - Navy, Class A technicians school appropriate to the maintenance task.
(c) Experience - Two (2) years of technical experience in addition to technical training".

Page 4, paragraph 3.3.3.2.2.6: Delete "a 2 mil thickness is permitted" and substitute "a 30 mil thickness is permitted provided reference designators and/or part identification remain clear, part replacement is not impeded, and compliance with MIL-E-5400 is maintained".

Page 4, paragraph 3.3.3.2.2.7: Delete "Government" and substitute "procuring activity".

Page 5, paragraph 3.3.3.2.2.9: Delete

Page 5, paragraph 3.3.3.2.2.11: Change "Air Filters" to "Service Access".

Page 5, paragraphs 3.3.3.2.2.13 and 3.3.3.2.2.14: Delete

Page 5, paragraph 3.3.3.2.2.16: Delete

Page 5, paragraph 3.3.3.2.3: Delete "/Manual Preparation".

Page 5, paragraph 3.3.3.2.3.1, last sentence: Delete

Page 6, paragraph 3.3.3.2.3.2: Delete "If feasible".

Page 6, second paragraph 3.3.3.2.3.2, after "diagnostic": Delete and substitute "interface specification".

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Page 6, paragraph 3.3.3.2.3.3: Delete

Page 6, paragraph 3.3.3.2.4.1, line one: Delete "shall" and substitute "should".

Page 6, paragraph 3.3.3.2.4.2: Add: "or a built-in voltmeter shall be provided".

Page 6, paragraph 3.3.3.2.4.3: Delete and substitute

"3.3.3.2.4.3 Chassis Test Points - Test points and built-in test facilities shall be included in the design of the equipment. Test points shall be chosen to provide a straightforward, logical step-by-step trouble shooting sequence, as well as provide a single end-to-end performance check".

Page 7, paragraph 3.3.3.2.4.4: Delete

Page 7, paragraph 3.3.3.2.6: Add "The meter shall indicate 0000 to 9999 hours".

Page 7, paragraphs 3.3.3.3, 3.3.3.3.1 and 3.3.3.3.1.1: Delete

Page 8, paragraph 3.3.4.1: Delete "MIL-E-5400 and 3.5.3.9" and substitute "WR-101, part I".

Page 8, paragraph 3.3.4.2: Delete "MIL-W-5088" and substitute "WR-101, part I".

Page 8, paragraph 3.3.6: Delete all after "WR-101, part I" and substitute "A control plan will be prepared in accordance with WR-101, part I". Delete all references to DOCUMENT and SUBMISSION DATE.

Page 9, paragraph 3.3.8: Delete "115 $\pm$ 1.0 V AC" and substitute "115 $\pm$ 3.0 V AC 400 cps $\pm$ 2%".

Page 9, paragraph 3.3.9.1: Delete "Curve IV" and substitute "Curve I".

Page 9, paragraph 3.3.11: Delete and substitute

"3.3.11 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as defined herein. The equipment shall operate within the bounds of Limits 2 and 3 of Figure 3 of MIL-STD-704A except that

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curve 3 of figure 3 shall not fall below 80 volts.

Operational malfunction may occur when the input voltage exceeds the above limits but remains within limits 1 and 4 of Figure 3 of MIL-STD-704A. However, no damage shall be suffered by the equipment and normal operation shall automatically be resumed upon return of the input voltage to levels within limits 2 and 3. The power required shall not exceed the following specified amounts:

AC Power (three phase) 115/200V, 400 cps Category "B", 185 VA".

Page 9, paragraph 3.3.11.1: Delete

Page 9, paragraph 3.4.1; Make the following changes:
First line: Delete "be designed to"
Third line: Delete "11" and substitute "12-14"
Sixth and Seventh lines: Delete "be designed to"
Ninth line: Delete "excited" and substitute "type"

Page 10, paragraph 3.4.2, line three: Delete "excited" and substitute "type".

Page 10, paragraph 3.4.3, line 1: Add: "output"; line 7: Delete "the least significant bit" and substitute "6 minutes". Add "The accuracy of a synchro to digital conversion shall be $\pm$ 6 minutes".

Page 10, paragraph 3.4.4: Delete "3.3.11.1" and substitute "3.3.11".

Page 10, paragraph 3.4.5, line 2: Delete "A-NEW central digital". Line 8: Delete "A-NEW digital".

Page 10, paragraph 3.4.6, after "following": Add "exclusive of mounting, control and connector protrusions": Also, change length to "30.75 inches"; height to "12.75 inches" and "width" to "10.75 inches".

Page 10, last line: Delete

Page 11, paragraph 3.5: Delete "A-NEW digital".

Page 11, paragraph 3.5.1, last line: Delete "11 bit" and substitute "12-14 bit".

Page 11, paragraph 3.5.1.1, line 9, after "5000 ohms": Add "the unbalance may be greater than a 2% unbalance. However, the effect due to this unbalance shall be no greater than a 2% unbalance when the converter presents a load of 5000 ohms".

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Page 12, paragraph 3.5.1.3: Delete "On each connector-----from a synchro transmitter". Add "The converter shall be capable of supplying 40 watts of rotor signal power".

Page 12, paragraph 3.5.1.4: Make the following changes:
Delete "11 bit" and substitute "12-14 bit"
Delete "per (4) and substitute "per (3)"
Delete "Output Data" and substitute "Input Data"
Delete "per (7)" and substitute "per (5)"

Page 13, paragraph 3.5.2.2, fifth sentence: Delete and substitute "Each output channel shall be capable of driving one control transformer having an impedance no less than 100 ohms, however, the total output power need not exceed that required to drive 16 440 ohm control transformers simultaneously".

Page 13, paragraph 3.5.2.3: Add "as per (6) of section 3.5.2.1".

Page 13, paragraph 3.5.2.4: Delete "Rate" and substitute "Requirement". Also add: "without the requirement that the digital data must be repeated".

Page 13, paragraph 3.5.3.2: Delete "A-NEW Avionics".

Page 14, paragraph 3.5.3.8: Delete first sentence.

Page 14, paragraph 3.5.3.9: Delete and substitute

"3.5.3.9 Electrical Connectors - External - Except for input power, the electrical connectors provided shall conform to MIL-C-81511. Power connectors shall comply with MIL-C-26482".

Page 14, paragraph 3.5.3.9.1: Delete

Page 15, paragraph 3.5.4.2: Delete "A-NEW Avionics".

Page 15: Add

"3.5.6 Manual Operation - A means shall be provided to manually enter an input binary word and a channel address. A display shall also be provided to visually monitor the output data request signal and a manual means of re-setting it. The output word shall also be monitored, with the exception of Test Sequence Identification Bits 22 through 27.

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3.5.7 Mounting - The equipment shall be suitable for hard mounting in the airplane".

Page 15, paragraph 4.1.1: Delete (4), (5) and (6).

Page 15, paragraph 4.1.2: Delete

Page 16, paragraph 4.2, first line: Delete "an".

Page 16, paragraph 4.2.1: Delete "ambient temperatures" and substitute "ambient and surface temperatures of temperature critical parts. This data shall be obtained under the extreme high temperature operating condition".

Page 16, paragraph 4.2.2: Delete and substitute

"4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all requirements of this specification and the contract. Preproduction tests shall include the following:

1. Environmental Tests in accordance with MIL-T-5422.
2. Interference Tests in accordance with WR-101, Part 1.
3. Shakedown Test
4. Power Input Test
5. Reliability Test
6. Maintainability Test
7. Acceptance Tests"

Page 16: Add

"4.2.2.1 Shakedown Test - Each preproduction equipment shall be subjected to a shakedown test of 100 continuous hours of failure-free operation. The tests shall be conducted in accordance with MIL-STD-781, using test level E. Failures encountered in the test will be the subject of detailed failure analysis and corrective action. The corrective action loop shall document the assignable cause of failure and the action taken. Corrective actions shall be approved by the procuring activity prior to implementation. The test will then be started from time zero. The performance characteristics to be measured shall consist of a series of functional tests, exercising all portions of the equipment in a manner as stringent as that required during its ultimate application".

Page 17, paragraph 4.4: Add "All inspection and test shall be under the supervision of the Government inspector". Delete "individual tests" and substitute "the following:

(1) Individual Test
(2) Special Test

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(3) Sampling Test
(4) Reliability Tests"

Page 18, paragraph 4.4.1.1, line 8: Delete "paragraph 4.6.7 cycle, phase 2" and substitute "Test Level E of MIL-STD-781". Delete subparagraph entitled "Shock". Delete first sentence of second paragraph starting with "The mechanical shock test ----". In third sentence of second paragraph delete ---- "paragraph 4.5.7 cycle, phase 2" and substitute "Test Level E of MIL-STD-781".

Page 19, paragraph 4.4.1.3.2(1): Delete "Figure 3" and substitute "Figure 5".

Page 19: Add paragraph 4.4.2

"4.4.2 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests".

Page 19: Add paragraph 4.4.2.1

"4.4.2.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity)".

Page 19: Add paragraph 4.4.2.2

"4.4.2.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests".

Page 19: Add paragraph 4.4.3

"4.4.3 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following:

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0#
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

Sampling Tests are not required when Reliability Assurance Tests are conducted.

#NOTE: If by contract action the Initial Production Test is deleted, a Sample Test shall also be conducted on one equipment from the first 10 produced".

Page 19: Add paragraph 4.4.3.1

"4.4.3.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run-in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment".

Page 19, paragraph 4.5: Delete "MIL-R-23094" and substitute "MIL-STD-781".

Page 19, paragraph 4.5.2: Delete and substitute

"4.5.2 Test Level - Test Level E of MIL-STD-781 shall be used with temperature cycling per Figure 1 of MIL-STD-781".

Page 19, paragraph 4.5.3: Delete and substitute

"4.5.3 Accept-Reject Criteria - Test Plan IV of MIL-STD-781 shall be used to determine the accept-reject criteria".

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Page 20, paragraph 4.5.4: Delete and substitute "All Equipment Test - Each equipment produced shall be tested for 100 hours. Prior to the 100 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extent the line as necessary to accommodate the data). These totals shall accumulate so that any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attn: Avionics Division".

Page 20, paragraph 4.5.5, line 6: Delete "13000" and substitute "4.9 times the MTBF". Lines 9 and 10: Delete "Test Level I, Table I, ------ of the equipment operating time".

Page 20, paragraph 4.5.7: Delete

Pages 21 thru 24, paragraphs 4.5.10 thru 4.5.16: Delete

Page 21: Add

"4.5.10 Non-Chargeable Failures - The contractor will receive approval to eliminate a failure from the reliability computation from the procuring activity when any of the following conditions apply:

(a) The contractor analytically and/or by test establishes the effectiveness of the corrective action (or change) to the satisfaction of the procuring activity and the corrective action or change is scheduled for introduction and/or retrofit by the contractor as approved by the procuring activity. Approval of any change by the procuring activity shall not protect any subsequent failure from classification.

(b) The contractor demonstrates to the satisfaction of the procuring activity that the failure was a dependent failure.

(c) The failure cannot be verified.

(d) Improper test or test equipment can be verified as the cause of failure to the satisfaction of the procuring activity.

(3) Approved preventive maintenance can be verified as being past due and the preventive maintenance action has been waived for the test by the procuring activity.

Chargeable Failures - All failures that are not classified as non-chargeable failures shall be considered chargeable failures.

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4.5.11 Final Reliability Report - The final reliability report shall be prepared in accordance with MIL-STD-781 and shall also include the following:

(1) Conditions to be avoided if a high reliability factor is to be maintained in field operation.

(2) Quality control procedures necessary to maintain a high standard of reliability for equipment produced in quantity".

Page 24, paragraph 4.6: Delete and substitute

"4.6 Maintainability Demonstration - A Maintainability Demonstration shall be conducted per Method I, Test Plan (b1) of Table II of MIL-STD-471".

Page 25, paragraph 4.6.1: Delete "in paragraph 3.3.3.2.3.3" and substitute "in paragraph 3.3.3.1".

Page 25, paragraph 4.6.5: Delete

Page 25, paragraph 4.7: Delete "preproduction tests, acceptance tests and life tests" and substitute "all tests".

Page 26: Add

"4.11 Test Data Reporting - MIL-STD-831 shall be used as a guide in the preparation of test reports".

Page 26, paragraph 6.1: Delete "11 bit" and substitute "12-14 bit".

Page 28, paragraph 6.9: Delete

Page 29, paragraph 6.11, under SPECIFICATIONS Military: Delete "MIL-W-5088; MIL-E-4682; MIL-I-6181; MIL-D-7018(WEP); MIL-R-23094(WEP) and MIL-C-38300" and add "MIL-C-81511 Connectors, Electric, Circular, High Density, Quick Disconnect, Environment Resisting, General Specification for".

Page 30, paragraph 6.11, under STANDARDS: Add

"MIL-STD-470 Maintainability Program Requirements (for Systems and Equipment)
MIL-STD-471 Maintainability Demonstration
MIL-STD-701 Preferred and Guidance Lists of Semiconductor Devices
MIL-STD-704 Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781 Test Levels and Accept/Reject Criteria for Reliability of Non-Expendable Electronic Equipment

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MIL-STD-785	Requirements for Reliability Program (For Systems and Equipments)
MIL-STD-831	Test Reports, Preparation of"

Page 30, paragraph 6.11: Delete "Weapons Specifications"; "WS-3099"; and "BUWEPS Instruction 4700.2A".

Page 30, paragraph 6.11: Add

"NAVAL AIR SYSTEMS COMMAND

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
WR-101	Weapons Requirements for Advanced ASW Electronic Systems Part I - Electromagnetic Control requirements for Advanced ASW Avionics Systems Part II - Selection and Testing of micro electric circuits for advanced ASW avionics systems; A-NEW".

Page 30, paragraph 6.11.1: Delete "procuring activity or as directed by the contracting officer". and substitute "Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120".

Page 30, paragraph 6.12: Delete

Figure 1, delete "11 bit" from description of signal flow between converter and computer.

Page 34: replace Figure 3 with newly revised Figure 3 of this amendment.

Page 35: replace Figure 4 with newly revised Figure 4 of this amendment.

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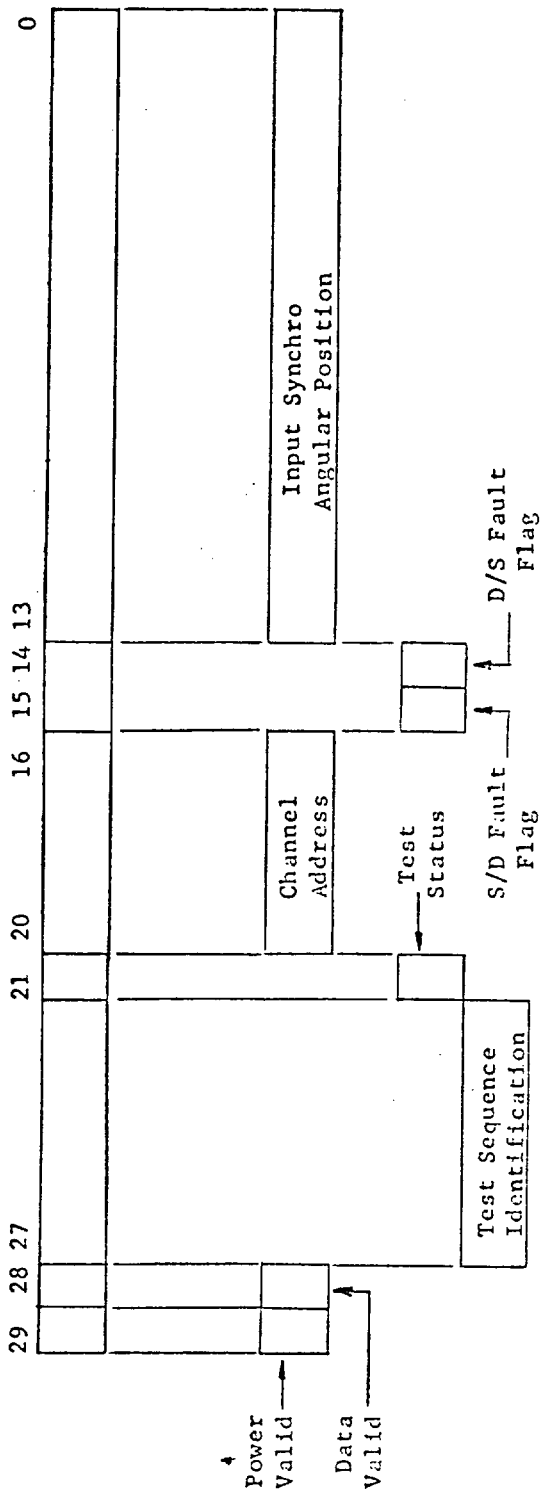


FIGURE 3 SYNCHRO ANGULAR POSITION WORD

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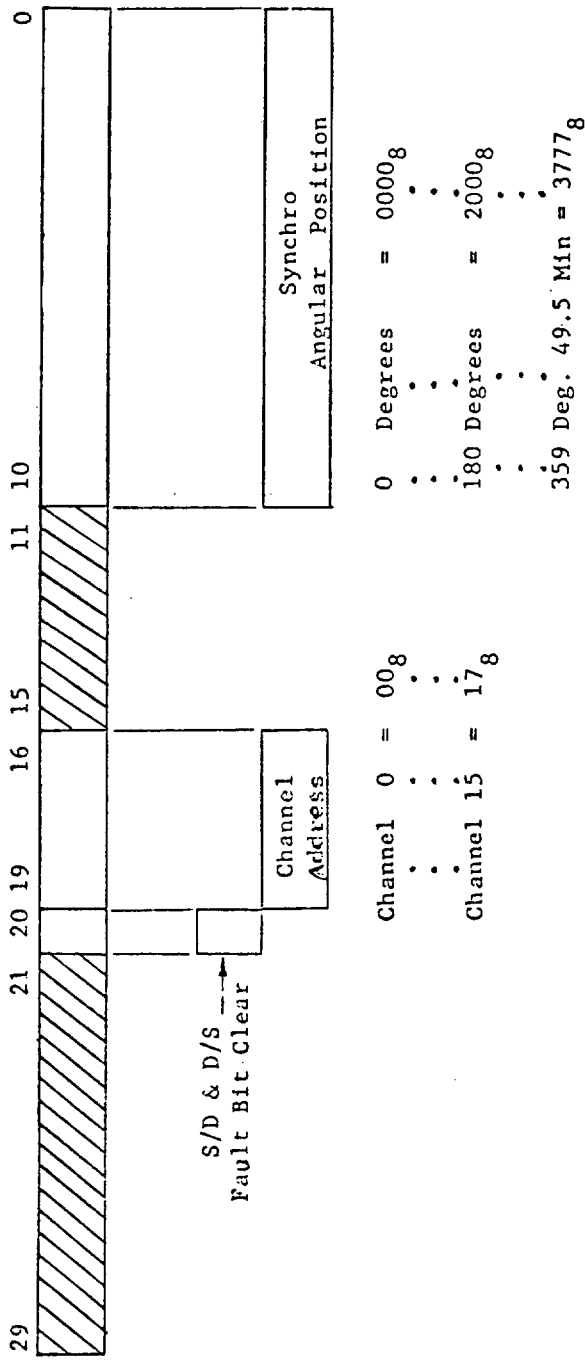


FIGURE 4 SYNCHRO ANGULAR POSITION AND CHANNEL ADDRESS WORD

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